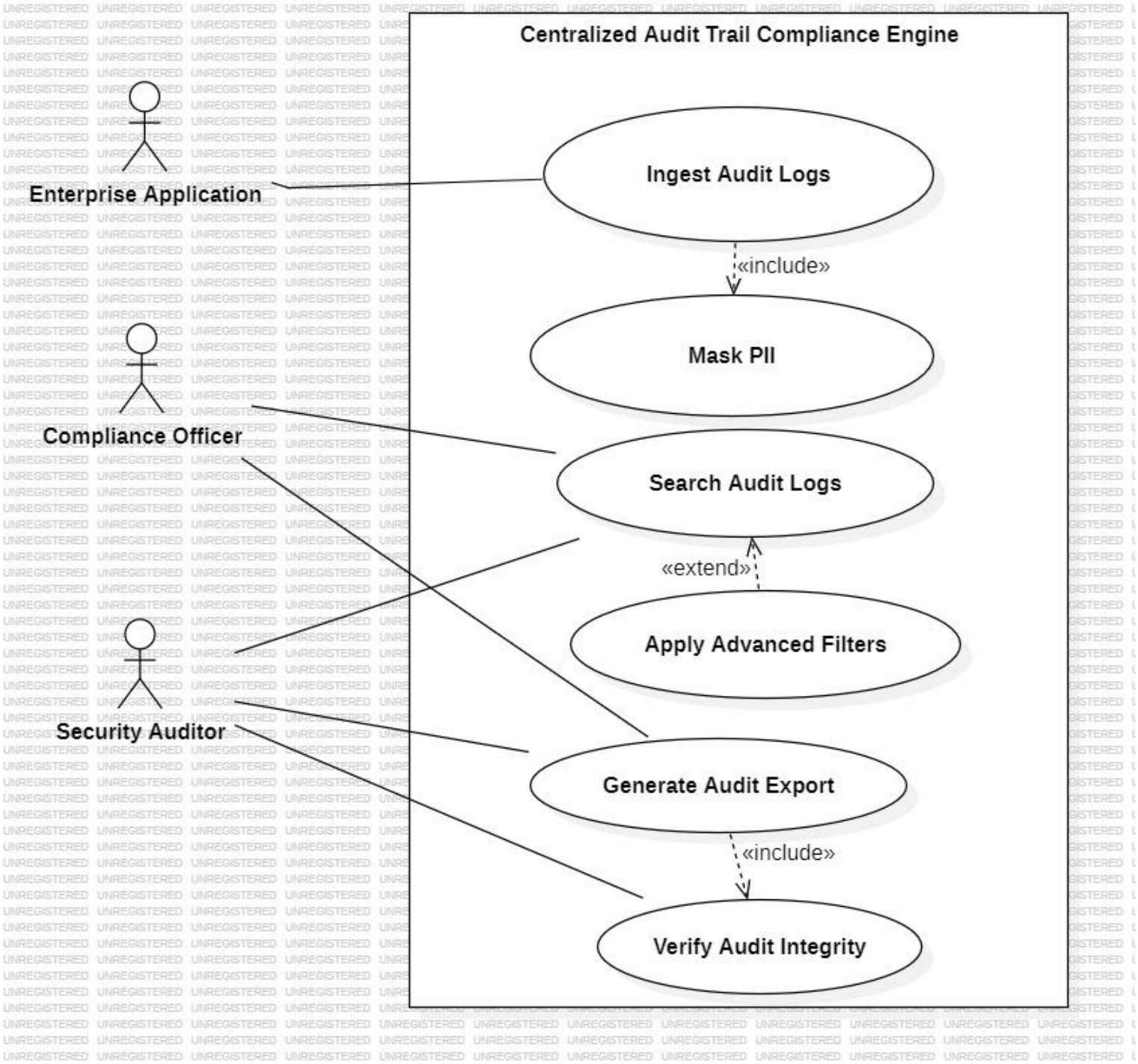
Requirements Table

Problem Statement #50 – Centralized Audit Trail Compliance Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse structured JSON audit logs, identify and mask PII fields such as Social Security Numbers and credit card numbers before persisting records to storage.	High	Pass: Masked audit log is saved with a SHA-256 integrity hash. Fail: Any cleartext credit card number or Social Security Number is stored.	Protects sensitive personal information and supports GDPR compliance.
FR-002	Functional	The system shall ingest audit event logs from enterprise applications and make them available for processing.	High	Pass: A valid audit log submitted by an enterprise application is successfully received and appears as a processable record. Fail: A valid submitted log is lost or rejected without an error.	Log ingestion is the foundation of centralized audit-trail collection.
FR-003	Functional	The system shall allow authorized users to search and filter stored audit records using specified criteria.	High	Pass: An authorized user provides valid search criteria and the system returns all matching audit records. Fail: Matching records are missing from the results.	Enables Compliance Officers and Security Auditors to investigate historical events.
FR-004	Functional	The system shall generate cryptographically verifiable immutable audit exports from selected audit records.	High	Pass: The generated export contains an integrity mechanism that successfully verifies the exported audit data. Fail: The generated export fails integrity verification or can be modified without detection.	Provides trustworthy audit evidence for compliance and security reviews.
FR-005	Functional	The system shall allow authorized users to verify the integrity of stored audit records and generated audit exports.	High	Pass: An unchanged record/export passes integrity verification, while a modified record/export is detected as invalid. Fail: A tampered record/export is reported as valid.	Helps auditors detect unauthorized modification of audit evidence.
NFR-001	Nonfunctional – Performance & Security	The system shall support complex filtering across 10 million historical audit records in under 1 second.	High	Pass: Benchmarking confirms that complex audit-log queries across 10 million records complete in less than 1 second under simulated peak load. Fail: The target latency is exceeded.	Ensures fast access to audit information even with a large historical dataset.
NFR-002	Nonfunctional – Security	The system shall ensure that sensitive PII is not stored in cleartext in persisted audit records.	High	Pass: Security testing confirms that all configured PII fields are masked before persistence. Fail: Any configured PII value is found in cleartext in stored audit records.	Prevents exposure of sensitive personal information and strengthens GDPR compliance.


Centralized Audit Trail Compliance Engine
Enterprise Application
Compliance Officer
Security Auditor
Ingest Audit Logs
«include»
Mask PII
Search Audit Logs
«extend»
Apply Advanced Filters
Generate Audit Export
«include»
Verify Audit Integrity

Use-Case Flow Specification

Centralized Audit Trail Compliance Engine

Use Case: Generate Audit Export

Primary Actors: Compliance Officer, Security Auditor

Goal: Generate a cryptographically verifiable and immutable audit export from selected audit records.

Preconditions

- The user is authorized to access audit records.
- Relevant audit records are available in the system.

Postconditions

- A cryptographically verifiable audit export is generated from the selected records.
- The generated export can be used as audit/compliance evidence.

Main Success Scenario

1. The Compliance Officer or Security Auditor selects “Generate Audit Export.”
2. The system requests the required selection criteria for the audit records.
3. The user specifies the required records or filtering criteria.
4. The system retrieves the matching audit records.
5. The system verifies the integrity of the selected audit records.
6. The system generates the audit export with cryptographic integrity information.
7. The system makes the generated export available to the user.
8. The user receives the audit export successfully.

Alternate Flow – Integrity Verification Fails

1. At Step 5, the system detects that one or more selected audit records fail integrity verification.

2. The system reports the affected records as invalid and does not treat them as trusted audit evidence.

3. The user is notified that the requested export could not be generated as a trusted export.

Related Use-Case Relationship

Generate Audit Export «include» Verify Audit Integrity.